

FreshFlow Beverages Pvt. Ltd.

FreshFlow Bottling Plant - Pune Production Facility
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STANDARD OPERATING PROCEDURE

Quality Inspection Procedure for Mixing Line 92

Department: Mixing

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Prepared by: Engineering / QA Department
Approved by: Plant Manager

Document Control Information

Parameter	Value
SOP Number	SOP-092
Title	Quality Inspection Procedure for Mixing Line 92
Department	Mixing
Plant	FreshFlow Bottling Plant - Pune Production Facility
Prepared By	Employee 4
Reviewed By	Anita Desai
Approved By	Rajesh Kulkarni - Plant Manager
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1. Purpose

Blend water, syrup, and CO2 to create a finished carbonated beverage product meeting Brix, pH, CO2, and microbiological specifications.

2. Scope

Mixing department operators.

3. Prerequisites and Requirements

3.1 Training Requirements

- Mixing and carbonation training
- CO2 safety training (SOP-SAF-001)

3.2 Tools and Materials Required

- Calibrated instruments as listed in the plant calibration schedule.
- Appropriate PPE as specified in Section 4.
- Completed permit-to-work (if required for maintenance activities).
- Access to relevant equipment manual and P&ID drawings.
- CMMS access to create and close work orders.

4. Health, Safety and Environmental Requirements

! WARNING

This SOP must not be carried out without appropriate PPE and, where required, a valid Permit to Work (SOP-SAF-006). If in doubt about safety, STOP and consult the Safety Officer (EMP009 / EMP017).

4.1 Identified Hazards

- CO2 (asphyxiation risk at concentrations >5% vol) - monitor GD101
- Cold product at 2-4°C - skin hazard and slip risk
- Pressurized systems (up to 6 bar CO2) - release risk

4.2 Emergency Response

In case of any emergency during this procedure: Press the nearest Emergency Stop, alert personnel in the vicinity, call the Safety Officer and Shift Supervisor. Refer to the Emergency Response Plan (SOP-SAF-004) and site evacuation procedure.

5. Detailed Procedure

5.1 Pre-Task Verification

1. Confirm the relevant equipment is available and no conflicting activities are planned.
2. Check shift handover log for any outstanding issues with this equipment.
3. Verify all required materials, tools, and chemicals are available and correctly labelled.
4. Obtain all required permits (PTW, hot work, confined space as applicable).
5. Brief all involved personnel on this SOP before starting.

5.2 Main Procedure

1. Isolate and LOTO the equipment per SOP-SAF-002. Verify zero energy state.
2. Allow equipment to cool to ambient temperature before touching internal parts.
3. Disassemble relevant sub-assembly according to the equipment manual section 6.
4. Inspect all removed parts. Photograph any damaged or worn components.
5. Clean the housing and contact surfaces. Remove all old gasket material.
6. Install new parts using food-grade lubricants where applicable. Do not over-torque fasteners.
7. Reassemble in reverse order. Verify all fasteners are tightened to specified torques.
8. Remove LOTO and test equipment at low speed/load before returning to production.
9. Verify key parameters (pressure, temperature, vibration) after restart.
10. Record the work in CMMS with parts used, cost, and technician details.

6. Quality and Verification Checks

Verification Check	When	Responsible	Record
All product contact surfaces visually clean	Before production	Operator	Shift Log
CIP conductivity <=20 uS/cm	After each CIP	Operator	CIP Record
CIP pH 6.5-7.5	After each CIP	Operator	CIP Record
Microbiological swab <10 CFU/cm2	After each CIP	QC Lab	Micro Report
Key process parameters in normal range	At startup	Operator	SCADA Historian
First-off product sample - Brix, pH, CO2	After startup	QC Analyst	QC Lab Report

7. Non-Conformance and Escalation

If any check fails or if the procedure cannot be completed as written: STOP the task. Do not force equipment to operate outside specification. Escalate to the Shift Supervisor (Suresh Patil / Ravi Deshpande). Raise a Non-Conformance Report (NCR) in the CMMS system. QC Manager (Anita Desai) must be notified for any product safety impact.

8. Records and Documentation

Record Type	Frequency	Responsible	Storage Location
Shift Log - Mixing	After each shift	Shift Supervisor	Shift Log Book / SCADA
CMMS Work Order	For every maintenance task	Technician	FreshFlow CMMS
CIP Record (FFB-QA-CIP-001)	After every CIP cycle	Operator	QA Filing System
Instrument Calibration Log	After calibration	QC/Maintenance	Calibration Database

A. Supplementary Notes

This SOP should be read in conjunction with the relevant equipment manual and the FreshFlow Beverages Good Manufacturing Practice (GMP) guidelines. Any suggested improvements to this SOP should be submitted to the Engineering Department using the Document Change Request (DCR) form FFB-ENG-DCR-001. This SOP is reviewed annually or whenever a significant process change occurs.

Related SOPs: SOP-SAF-002 (LOTO), SOP-GEN-010 (GMP), SOP-QA-005 (HACCP), SOP-GEN-001 (Shift Handover), SOP-GEN-006 (Plant Startup). For questions on this procedure, contact the Engineering or QA Department.